Dipannoy Das Gupta

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Website | Google Scholar | LinkedIn | GitHub

HIGHLIGHTS

AI Researcher with 3+ years of experience in predictive and generative modeling in materials science. Expertise encompasses graph neural networks, transformers, inverse design, and state-of-the-art AI architectures. Deeply focused on developing physics-informed machine learning models to accelerate materials discovery.

RESEARCH EXPERIENCE

Graduate Research Assistant, University of South Carolina — Aug 2022 – Present

- **Single-Step Crystal Structure Prediction:** Developed a conditional generative model for inorganic crystals that predicts highly accurate structures significantly faster than state-of-the-art baselines (CDVAE, CrystalFlow, and DiffCSP). The model successfully learns complex symmetry operations by integrating multiple layers of physical priors, generating structures 17× faster than CrystalFlow while improving overall accuracy.
- **Multi-Property Inverse Design:** Designed a joint predictive model to screen crystal libraries for materials balancing conflicting properties (formation energy, band gap, and stiffness). Training all three jointly recovers 60% of the true optimal trade-off surface, compared to a 50% recovery rate from separately trained single-property models. The model identifies the top 2% non-dominated crystals from a 4,800-structure library, outperforming random screening by 35% with every proposal traceable to a real material.
- **Adaptive Edge Pruning for Crystal Property Prediction:** Augmented Graph Neural Networks with an edge-scoring network designed to dynamically prune chemically irrelevant edges during message passing. This adaptive scheme reduces band gap prediction error by up to 2.8% and achieves similar or improved formation energy predictions while retaining only 62%–68% of crystal graph edges. The resulting pruned graphs maintain greater representational diversity across nodes, minimizing over-smoothing.
- **Domain Adaptation for Material Screening:** Implemented parameter-based domain adaptation techniques across state-of-the-art models (SchNet, PAiNN, ALIGNN) to overcome data scarcity challenges. This transfer learning approach improved predictive accuracy by **1.5–2**× on unseen chemical spaces. Deployed the refined model to screen 222,960 halide perovskites, yielding 40 previously unreported photovoltaic candidates.

PUBLICATIONS & PREPRINTS

- Dey, Sourin, **Dipannoy Das Gupta**, Lai Wei, Sadman Sadeed Omee, and Jianjun Hu. "uFlowCSP: Crystal Structure Prediction using Mean flow generative models." *arXiv preprint arXiv:2609.09799* (2026).
- **Gupta, Dipannoy Das**, Zachary JL Bare, Suxuen Yew, Santosh Adhikari, Brian DeCost, Qi Zhang, Charles Musgrave, and Christopher Sutton. "Leveraging domain adaptation for accurate machine learning predictions of new halide perovskites." *arXiv preprint arXiv:2401.10998* (2024).
- *Manuscript in preparation - preprints expected October 2026:* "Inverse Design of Crystalline Materials via Multi-Property Screening Model"; "Cooperative Graph Neural Networks: Learned Edge Pruning for Crystal Property Prediction".

EDUCATION

PhD Candidate, Department of Computer Science, University of South Carolina — 2025 – Present

M.Sc. in Computer Science, University of South Carolina — 2022 – 2024

B.Sc. in Computer Science & Engineering, Bangladesh University of Engineering & Technology (BUET) — 2015 – 2019

SKILLS

- **Research:** Graph Neural Networks (GNN), Equivariant Networks, Predictive Modeling, Diffusion & Generative Models, Energy-Based Models, Inverse Design, Domain Adaptation, Multi-Objective Optimization, Atomistic Simulation (Pymatgen, ASE, VASP), Self-supervised Learning
- **ML Libraries:** PyTorch, PyTorch Geometric, DGL, TensorFlow, Keras, Scikit-Learn, Weights & Biases (W&B)
- **Languages & Systems:** Python, C#, C++, JavaScript, Node.js, React, ASP.Net Core, SQL; Git, Linux, High Performance Computing (HPC), Slurm, CUDA